

WORK EXPERIENCE

Curriculum Developer & Instructor

Masterschool | Aug 2026 – Present | Berlin, Germany · Freelance · Remote

Develop and teach hands-on courses in ML and AI application engineering, covering RAG, MCP, tool calling, structured outputs, and agentic orchestration.

- Design and build end-to-end agentic AI applications using LangGraph, with retrieval, MCP-based tool integration, validation, and multi-step orchestration.

Project Manager – AI & Pattern Recognition

MyGroove GmbH | Mar 2025 – Jun 2026 | Salzburg, Austria · Full-time · On-site

Delivered production AI workflows and ML-powered features for a music learning platform, working across product and engineering to balance model quality, latency, cost, and reliability.

- Built and scaled a GenAI localization pipeline for 20k+ minutes of educational content, orchestrating translation, voice synthesis, lip-sync, reviewer approval, and delivery; reduced dubbing costs by 67% and enabled the English product launch.
- Deployed an AWS-based RAG QA system that grounded LLM validation in source transcripts, approved terminology and lesson context, reducing manual review effort by 73%.
- Improved production audio-recognition and performance-scoring systems by diagnosing learner-session failures across model inference, annotation tooling, and app integration.
- Reduced quiet-input failures by 43% by tuning an RMS-envelope-based automatic gain-control system without clipping transients or amplifying background noise.

Data Science Consultant

PhonicScore GmbH | Mar 2023 – Dec 2024 | Mumbai, India · Freelance · Remote

Led technical improvements to client-specific audio-event detection systems across model development, evaluation, and production inference.

- Trained and evaluated Conv-LSTM models on multi-GPU infrastructure, achieving a 51% relative improvement in F1 over the baseline through targeted dataset changes, fine-tuning, and analysis of onset, timing, and false-positive errors.
- Developed a C++ and TensorFlow Lite inference harness for low-latency execution on iOS and Android, integrating temporal alignment, threshold calibration and edge-case handling.
- Built audio augmentation and validation workflows addressing noise, loudness, room acoustics and recording-level variation to improve robustness under real-world conditions.

Junior Data Engineer

Learnfield GmbH | Jul 2022 – Feb 2023 | Berlin, Germany · Full-time · On-site

- Developed data pipelines and experimentation tooling for supporting the training, evaluation, and optimization of piano-transcription models.
- Maintained containerized ML infrastructure on AWS and adapted experimental models for production and mobile deployment under strict latency, compute, and memory constraints.

Master's Thesis Intern

Learnfield GmbH | Oct 2021 – Jul 2022 | Berlin, Germany · Full-time · Remote

- Researched compression techniques for transformer-based piano-transcription models, reducing model size by 4.8× through knowledge distillation, pruning, and quantization while targeting resource-constrained mobile inference.

Student Research Assistant

OvGU AILab | Mar 2019 – Jun 2021 | Magdeburg, Germany · Part-time · On-site

- Continuously improved the ISMIR Paper Explorer using BERT embeddings and semantic similarity to enable interactive research discovery and trend exploration.
- Developed Python-based AI research prototypes for music generation, data processing, and model evaluation, including demos presented at the AI Song Contest 2020.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL, TypeScript, Bash
- **ML & Model Development:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers; distributed training, fine-tuning, model evaluation, quantization, pruning, knowledge distillation
- **Generative AI & Agent Systems:** LLMs, RAG, tool calling, MCP, structured outputs, agentic workflows; SFT, LoRA/QLoRA, DPO; LangGraph, AWS Bedrock
- **ML Systems & Inference:** AWS SageMaker, vLLM, ONNX, MLflow; model serving, distributed inference, experiment tracking, observability, latency/throughput/cost optimization
- **Backend, Data & Cloud Infrastructure:** FastAPI, gRPC, PostgreSQL, pgvector, Redis, Temporal, Docker, Kubernetes, Terraform, GitHub Actions; AWS EC2, ECS/EKS, Lambda, S3, SQS, CloudWatch

PUBLICATIONS

Towards Compressed Transformers for Piano Transcription

Thomas Prätzlich, Richhiey Thomas, Sebastian Stober.

In Proceedings of *DAGA 2026 – 52nd Annual Conference on Acoustics*, Dresden, Germany.

EDUCATION

M.Sc. in Data and Knowledge Engineering

Otto-von-Guericke University of Magdeburg, Germany, 2018 - 2022

Bachelor of Engineering in Computer Engineering

University of Mumbai, India, 2013 - 2017